Yonwoo Choi

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Education

Seoul National University *Mar 2022 – Feb 2024*
M.S in Computer Science and Engineering

- **Advisor:** Hanbyul Joo
- **Thesis:** A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images
- **GPA: 3.73/4.3**
- Coursework: 3D Computer Vision, Multimodal Deep Learning, Advanced Graphics, Advanced Data Mining, Mobile Computing

University of Virginia *Aug 2019 – May 2021*
B.A in Computer Science

- **GPA: 3.77/4.0**
- **Coursework:** Data Structures, Algorithms, Computer Architecture, Web Applications, Software Testing, Human Computer Interaction

Research Interest

My research lies at the intersection of 3D computer vision and generative AI, with a focus on creating realistic synthetic data for computer vision applications. My work explores the use of generative models to create high-quality synthetic training data, bridging the gap between simulation and real-world performance. I am passionate about developing practical systems that can capture, understand, and reconstruct complex 3D scenes and objects.

Publications

SVAD: From Single Image to 3D Avatar via Synthetic Data Generation with Video Diffusion and Data Augmentation **CVPR 2025 Workshop**

Yonwoo Choi

A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images **M.S Thesis**

Yonwoo Choi, Hanbyul Joo

Work Experience

GenGenAI *Seoul, South Korea*
Algorithm Engineer *Sept 2025 – Present*

- Sim2Real with 3D Simulation
- Video Diffusion Models for Synthetic Data Generation

SECERN AI *Seoul, South Korea*
Research Engineer *Mar 2024 – Sept 2025*

- Animatable 3D Human Avatars

Visual Computing Lab *Seoul, South Korea*
Graduate Researcher *Nov 2021 – Feb 2024*

- 3D Egocentric Hand Pose Estimation
- Multiview Capture System for Human Pose Estimation

Open Source

Single Image to 3D Human Avatar *May 2025*
SECERN AI

- Official Implementation of the paper SVAD
- Synthetic Data Generation Pipeline for 3DGS Avatar Training

Facebuilder *Mar 2025*
SECERN AI

- 3D Face Mesh Deformation with Pin-Based Interaction
- Multiview Face Modeling with Real-time 3D Consistency

SMPL Fitting *Jun 2024*
SECERN AI

- Fitting SMPL Parameters to 3D Human Joints
- Parametric Human Shape and Pose Optimization

Egocentric Human Pose Estimator *Dec 2023*
Visual Computing Lab

- Unofficial Implementation of the Paper xR-EgoPose
- Egocentric 3D Human Pose Estimation from a Single Fisheye Camera

Multiview Human Pose Capture System *Apr 2022*
Visual Computing Lab

- 5 GoPro Camera Setup for 3D Human Reconstruction
- Camera Calibration with Zhang's Method

Teaching

Teaching Assistant, Seoul National University *Fall 2023*
4190.408 001 - Artificial Intelligence (Instructor: Hanbyul Joo)

Teaching Assistant, Seoul National University *Spring 2022*
4190.773 002 - 3D Computer Vision (Instructor: Hanbyul Joo)